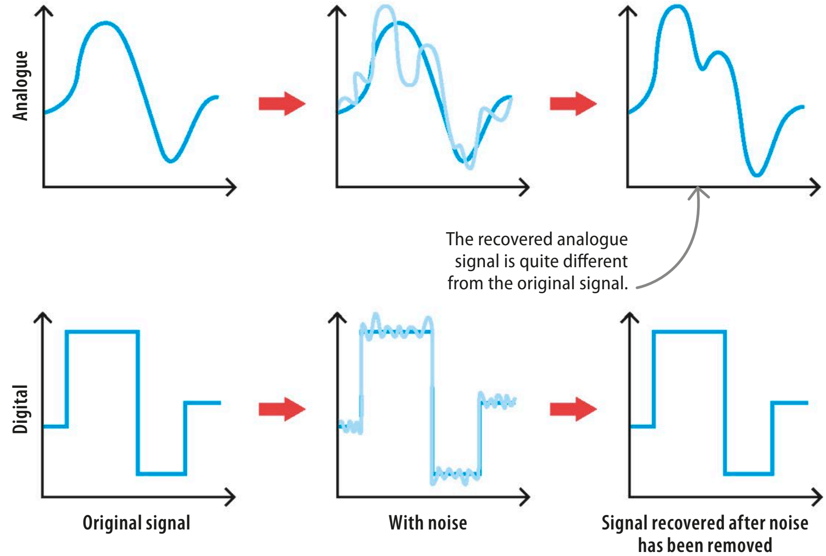
Noise

Electronic signals are rarely exact. Imperfections in the original medium or thermal (heat), electric, or solar interference is called noise, and can cause the signal to fluctuate. If the signal "1" is sent over a wire, it can be received as any value between 0.75 and 1.25, so the sender can never be assured that the signal originally sent is the signal received. Digital signals work in step-like increments, which means they are closer to the original signal and easier to receive. Analogue circuits, however, work in tiny, smooth increments, so some precision is always lost to noise.

▷ **Signal-to-noise ratio**

Noise adds random extra information to an analogue signal, making the signal less and less like the original signal. In contrast, the differences in the "on" and "off" states are so great in digital signals that it is easy to work out what the original signal was, even though there is usually some noise.



Pros and cons

Analogue computers are built at the hardware level; each computer is designed for a specific task, which makes them very accurate. However, since they're so specialized they can't be easily reprogrammed to carry out new tasks. Changing a program takes a lot of manual effort and might even require buying new components. Digital computers are more flexible and can be programmed to do unlimited tasks. Writing a new code in a digital system is easier than redesigning a motherboard.

Analogue computers assure accuracy and precision.

These allow real-time operation and simultaneous computation.

Analogue computers often consume less power and execute some tasks faster.

Analogue signals are a natural way of storing data. There is no quantization noise.

TOP TECH

Hybrids

Hybrid computers can combine the speed of analogue programming with the accuracy of digital programming to get the best of both worlds. So far, hybrids aren't widespread beyond specialized fields, such as radar systems and scientific calculations. However, after being unused for years, analogue computers are starting to make a comeback in programming.